MIL-P-46428 (Ord)
16 November 1960

MILITARY SPECIFICATION

PRIMER, ELECTRIC, T104E9
LOADING, ASSEMBLING, AND PACKING
(INCLUDING LOADING OF ELECTRIC IGNITION ELEMENT T9E6)

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer designated as Primer, Electric, T104E9, for use in artillery ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN-P-223 - Powder, Black
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

FSC 1390

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DRAWINGS

ORDNANCE CORPS

DXP-108343 - Primer, Electric, T104E9, Loading Assembly
F-8799363 - Box, Set-up, Ammunition for Primer, Electric, T104E9
F-8799364 - Box, Packing, Ammunition for Primer, Electric, T104E9

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-(See 6.4) - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primer.-The loaded primer shall comply with all requirements specified on Drawing (dwg.) DXP-108343, and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading, shall not exceed 0.5 percent when determined as specified in 4.4.1.

3.4 Non-functioning.-The head assemblies shall not function when tested as specified in 4.4.2.

3.5 Functioning.-The head assemblies shall function within 3 milliseconds, maximum, when tested as specified in 4.4.3.

3.6 Proving ground.-The primer shall comply with the following requirements when tested as specified in 4.4.4.

a. The primer shall not fail to fire.
b. The primer assembly shall not rupture or the parts become detached.
c. There shall be no gas blow back.

3.7 Workmanship.-All parts shall be free of chips, burrs, grease, rust, dirt and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.3 and 4.4 (except proving ground tests), and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

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e. Certificates of analyses on all material procured directly by the contractor, when such material is covered by referenced Government specification.

f. Number of items in the lot.

g. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawing and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records, Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operation, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

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4.2 Initial production.-The first lot produced in accordance with 4.3.1 and all successive lots, prior to Government acceptance of the initial production quantity, shall be inspected as required by this specification, except that the sampling plans for non-functioning, functioning and firing shall be as specified below. The contractor shall continue this plan of inspection until three consecutive lots have complied with the requirements.

4.2.1 Non-functioning (see 3.4) Critical defect.-A sample of 150 head assemblies shall be selected from each lot for this test. If any assembly fails to comply with the applicable requirement, the lot shall be rejected. Assemblies used for this test shall not be returned to the lot. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.2.2 Functioning (see 3.5) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IV A, code letter N and an AQL of 0.15 percent. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

4.2.3 Firing (Proving ground, see 3.6) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IV A, code letter K and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.4.

4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of primer assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Primer parts of one interfix lot number from one manufacturer.

b. Black powder of one lot number from one manufacturer.

c. Ignition element components of one interfix lot number from one manufacturer.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Electric ignition element, prior to pressing charge (see dwg. CXP-108345).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Weight of charge	Balance
102.	Foreign matter in charge or in charge cavity	Visual
Minor:	None defined	

4.3.2.2 Electric ignition element, prior to inserting cup support (see dwg. CXP-108345).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Charge not pressed	Visual-Manual
102.	Disc missing or mutilated	Visual
Minor:	None defined	

4.3.2.3 Electric ignition element, (see dwg. CXP-108345).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Total length	Gage
102.	Support cup missing, mutilated or not fully seated	Visual
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

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4.3.2.4 Head assembly (see dwg. DXP-108344).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Depth to electrode	Gage
102.	Retainer loose or not fully seated	Manual
103.	Flash hole completely obstructed	Visual
Minor:	AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.5 Assembly, prior to inserting diaphragm (see dwg. DXP-108343).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Weight of black powder charge	Balance
Minor:	AQL 1.00 percent	
201.	Foreign matter in charge	Visual
202.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.6 Assembly, after inserting first diaphragm (see dwg. DXP-108343).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Diaphragm missing, punctured or not sealed 360°	Visual
Minor:	AQL 1.50 percent	
201.	Location of diaphragm	Gage
202.	Diaphragm inverted	Visual
203.	Evidence of poor workmanship (see 3.7)	Visual

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4.3.2.7 Assembly (see dwg. DXP-108343).

Categories	Defects	Method of inspection
Critical:		
1.	Split or cracked body or head	Visual
2.	Flash hole(s) missing	Gage
Major:	AQL 0.40 percent	
101.	Head loose to finger torque	Manual
102.	Liner punctured (as seen through flash holes)	Visual
103.	Threads not coated with acid proof black paint	Visual
* 104.	Any flash hole not completely sealed	Visual
Minor:	AQL 2.50 percent	
201.	Total length, maximum (max.)	Gage
202.	End diaphragm above flush	Visual
203.	Threads damaged	Visual-Gage
204.	End diaphragm inverted or not sealed 360°	Visual
205.	Marking misleading or unidentifiable	Visual
206.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.8 Sealed, barrier bag (see dwg. F-8799363).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Seal improper or incomplete	Visual
102.	Bag punctured, torn or cut	Visual
Minor:	None defined	

* Sketches 1, 2 and 3 on Figure 1 indicate acceptable primers. Furthermore, a primer with holes displaying the conditions 4, 5 and 6 may be classed acceptable provided that the visible liner is thoroughly impregnated with a paint or lacquer and not more than 20 percent of the holes have this condition.

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4.3.2.9 Sealed set-up box (see dwg. F-8799363).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual
102.	Sealing strip incomplete, badly wrinkled or not in contact with surface of box for at least 1/2 inch around entire box	Visual
Minor:	AQL 0.40 percent	
201.	Contents move upon shaking	Manual

4.3.2.10 Box, packing (see dwg. F-8799364).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Hardware or strapping missing, broken or loose	Visual-Manual
102.	Board broken or split	Visual
103.	Contents exposed	Visual
Minor:	AQL 1.00 percent	
201.	Contents move when shaken	Manual
202.	Marking incorrect, misleading or unidentifiable	Visual

4.3.3 Normal production.-After acceptance of the initial production lots, testing shall be as follows:

4.3.3.1 Moisture content of black powder (see 3.3) Major defect.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test for verification at least one completed primer assembly from each eight hours production. Composite samples shall not be used in determining compliance. The black powder shall be tested as specified in 4.4.1 and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.3.3.2 Non-functioning (see 3.4) Critical defect.-A sample of 50 head assemblies shall be selected from each lot for this test. If any assembly fails to comply with the applicable requirement, the lot shall be rejected. Assemblies used for this test shall not be returned to the lot. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

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4.3.3.3 Functioning (see 3.5) Major defect.-A sample of 125 head assemblies shall be selected at random from each lot for this test. If two or more assemblies fail to comply with the requirement specified, the lot shall be rejected. If only one defective is found, a second sample of 250 assemblies shall be tested. If the combined number of defectives in both the first and second samples is two or more, the lot shall be rejected. Test equipment shall be in accordance with 4.3.4.

4.3.3.4 Firing (Proving ground) (see 3.6) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, using Table IV-A, code letter H and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.4.

4.3.3.5 Electrical resistance (see dwg. XP-108344) Critical defect.-This test shall be conducted 100 percent. Any head assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.5, using equipment in accordance with 4.3.4.

4.3.3.6 Grade of black powder (see dwg. DXP-108343) Major defect.-The contractor shall select approximately 100 grams of black powder from each container for this test. If any sample fails to comply with the granulation requirement, the container from which it was taken shall not be used in the loading of the primers. The grade of black powder shall be determined as specified in Specification JAN-P-223.

4.3.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.3 and 4.4, when applicable, the contractor shall have available and utilize correctly, all necessary gages, measuring and testing equipment (see 6.4). When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.4 Test methods and procedures.

4.4.1 Determination of moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification JAN-P-223.

4.4.2 Non-functioning.-The head assemblies shall be tested for non-functioning by subjecting them to a 10 volt discharge from a two microfarad condenser, using the electrode for one contact and the outer surface of the head as the other.

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4.4.3 Functioning.-The head assemblies shall be tested for functioning by subjecting them to a 160 volt discharge from a two microfarad condenser, using the electrode for one contact and the outer surface of the head as the other.

4.4.4 Proving ground.-This test shall be performed at a Government proving ground. The sample primers shall be assembled as a part of the complete round for which the primer was designed and shall be subjected to a rapid fire test in a gun for which the round is standard. Observation shall be made for compliance with the applicable requirements.

4.4.5 Electrical resistance.-The resistance shall be measured using an electrical circuit which limits the current and voltage applied to the head assembly to .060 milliamperes max. and 10 millivolts max. with the electrode as one contact and the primer head as the other.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The primer shall be preserved and packaged in accordance with dwg. F-8799363.

5.2 Packing.

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. F-8799364.

5.3 Marking.-Marking shall be in accordance with dwg. F-8799364.

5.4 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Check test reimbursement.-The contracting officer shall arrange to have the contractor reimbursed for the expense incurred in the performance of the check test.

6.3 Check test for possible deterioration.-The following tests are designed to prevent loading major ammunition items with primers which have deteriorated. The tests should be performed on each lot of artillery primers which have been stored for more than two years or which have been subjected to any adverse conditions, however brief, within that period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check tests in the contract.

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6.3.1 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The black powder shall be removed from the flash tube and the moisture content determined[1] in accordance with the method specified in Specification JAN-P-223. If the moisture content of any of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected.

6.3.2 Non-functioning.-The number of samples, acceptance and rejection, shall be as specified under the sampling plan designated by the code letter H, AQL of 0.40 percent of Standard MIL-STD-105. The samples shall be tested as specified in 4.4.2.

6.3.3 Functioning (see 3.5).-The sampling plan for the test shall be in accordance with Table II. Any head assembly which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.3, except that the flash tube and the head assembly shall be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable.

TABLE II

Sample	Sample Size	Size	Combined Samples Acceptance Number	Combined Samples Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

* No acceptance on this sample.

6.3.4 Electrical resistance.-This test shall be conducted 100 percent on the entire lot. Any head assembly which fails to comply with the applicable requirement on dwg. DXP-108344, shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.5, using equipment in accordance with 4.3.4.

6.4 Inspection equipment.-Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, to determine availability of inspection equipment designs.

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[1] Since black powder is hygroscopic, it is advisable to place the powder in a dry air-tight container immediately upon removal, and to run the determination as soon as possible thereafter.

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Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing activity:
Army-Ordnance Corps

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ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

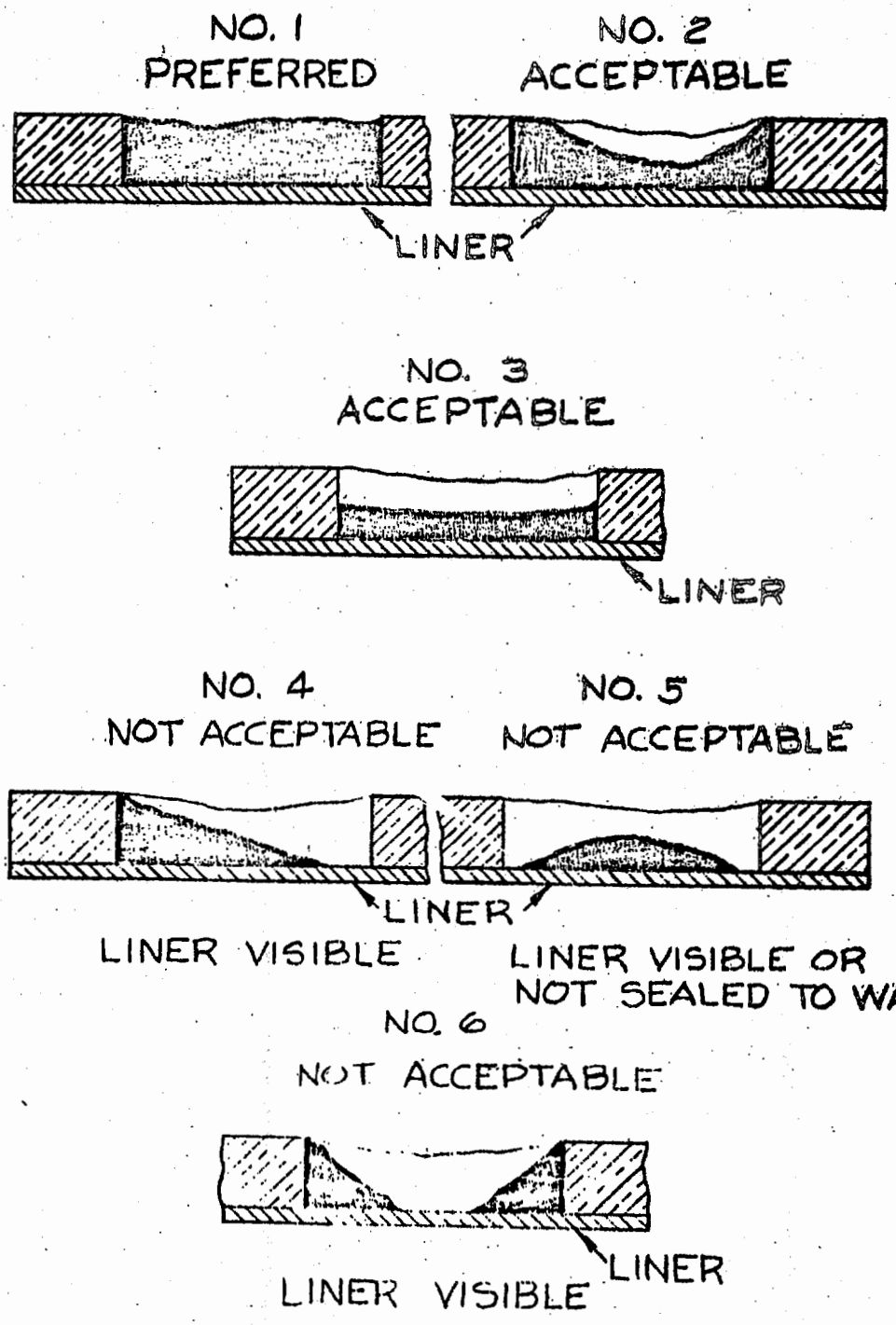


FIGURE 1

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